KING MONGKUT'S UNIVERSITY OF TECHNOLOGY NORTH BANGKOK
Faculty of Applied Science / Faculty of Information Technology
FINAL EXAMINATION — SEMESTER 1, ACADEMIC YEAR 2026 (PRACTICE SET B)

Course: 060243106 / 060223119 Data Structures and Algorithms | Time: 3 Hours (09:00 - 12:00)
Total Marks: 70 Marks (Normalized to 35% of Final Grade) | Lecturer: Assoc. Prof. Dr. Pradit Pitaksatheinkul

STUDENT INFO: Name: ..... ID: ..... Room: ..... Seat: .....
RULES: Open Book (Textbooks & Notes Allowed) | Scientific Calculator Allowed | NO Mobile Phones/Smart Devices

Question 1: Hashing & Open Addressing with Double Hashing (10 Marks)

Given an empty Open Addressing Hash Table with TableSize = 13 (indices 0 to 12).
Keys are mapped by primary hash function: $h_0(x) = x \bmod 13$.
Collisions are resolved by Double Hashing: $h_i(x) = (h_0(x) + f(i)) \bmod 13$, where $f(i) = i \times (11 - (x \bmod 11))$.
Insert the following 6 keys sequentially: [43, 17, 82, 95, 35, 160].

Index	0	1	2	3	4	5	6	7	8	9	10	11	12
Key													
Info													

1.1 Step-by-Step Collision & Probing Trace Table: [4 Marks]

Key (x)	$h_0(x) = x \bmod 13$	Double Hashing Step & Calculation f(i)	Final Index	Collisions
43				
17				
82				
95				
35				
160				

1.2 Collision Analysis: Which key collided the most? State the key, total collisions, and indices attempted: [1.5 Marks]
- Key: ..... Collisions: ..... Indices Attempted: .....

1.3 Total Collision Count: Total number of collisions and resolving probes accumulated: ..... times [1.5 Marks]
(Warning: Count cumulative probes only. Do NOT double count collision vs resolve)

1.4 Load Factor Percentage: Calculate the exact Load Factor percentage after inserting all 6 keys: [1 Mark]
- Formula & Calculation: $\lambda = (N \times 100) / \text{TableSize}$ = ... = ... %

1.5 Rehashing Threshold & New Size: If Rehashing occurs at 70% capacity: [1 Mark]
- How many more keys must be inserted to trigger Rehashing? .....
- What is the TableSize of the newly created hash table? .....

1.6 Concept of Wrap-Around: Define the concept of 'Wrap-Around' in Hashing as taught in class: [1 Mark]
- Definition: .....

Question 2: Priority Queue (Binary Min-Heap) Operations & Array Tracing (10 Marks)

Given the sequence of 10 integer keys: [65, 30, 48, 15, 22, 10, 50, 18, 25, 8].
Trace the 1-based array representation for each of the four operations below (Index 0 is unused / None).

2.1 Array State 1: After sequentially inserting all 10 elements into Min-Heap (Percolate Up): [3 Marks]

Index	0	1	2	3	4	5	6	7	8	9	10	11
Heap Array	None											

2.2 Array State 2: After executing deleteMin() operation #1 (Percolate Down): [2 Marks]

Index	0	1	2	3	4	5	6	7	8	9	10	11
After Del 1	None											

2.3 Array State 3: After executing insert(12) (Percolate Up): [2 Marks]

Index	0	1	2	3	4	5	6	7	8	9	10	11
After Ins 12	None											

2.4 Array State 4: After executing second deleteMin() operation #2 (Percolate Down): [1 Mark]

Index	0	1	2	3	4	5	6	7	8	9	10	11
After Del 2	None											

2.5 Binary Heap 1D Array Mathematical Formulas: [1 Mark]
For a node at index i in a 1-based array implementation:
- Left Child Index: ..... | Right Child Index: ..... | Parent Index: .....

2.6 Truncation Rule: Why must integer floor division (e.g., Python //) be used for parent calculation, and why is rounding up prohibited? [0.5 Marks]
- Explanation: .....

2.7 Percolate Down Code Condition: State the exact Python if-else condition used to pick the smaller child before swapping: [0.5 Marks]
- Code: .....

Question 3: Insertion Sort Tracing & Position Moves (10 Marks)

3.1 Worst-Case Mathematical Analysis (N = 10 Elements): [3 Marks]

- Define Worst-Case input configuration: .....
- Total Position Moves Formula & Exact Calculation: Total Moves = ... = ... moves.
- Best-Case Position Moves: ..... moves.

3.2 Pass Tracing on 10-Element Array: [4 Marks]
Given array: [54, 38, 16, 27, 10, 62, 45, 9, 70, 31].

Pass (p)	Key Element	Elements Shifted (Position Moves)	Array State After Pass p
p = 3	27		
p = 6	45		

3.3 Program Modification: How do we modify Insertion Sort to count total position moves in code? [3 Marks]

- Describe where the move counter increment statement must be placed:

.....

Question 4: Selection Sort Tracing (Test Program 2 Worksheet Template) (10 Marks)

Given the 7-element unsorted array: [64, 25, 12, 38, 11, 49, 20].
Write the exact steps for Round 1, Round 2, Round 3, Round 4 according to the classroom format: [8 Marks]

Round 1:

1. Find minimum: From unsorted array [64, 25, 12, 38, 11, 49, 20], the minimum is ..... at index .....
2. Swap: swap minimum with first element of unsorted array (64)
   - Before swap: [.....]
   - After swap: [.....]
3. Result: Now 1 element sorted. The remaining unsorted array is: [.....]

Round 2:

1. Find minimum: From unsorted array [.....], the minimum is ..... at index .....
2. Swap: swap minimum with first element of unsorted array
   - Before swap: [.....]
   - After swap: [.....]
3. Result: Now 2 elements sorted. The remaining unsorted array is: [.....]

Round 3:

1. Find minimum: From unsorted array [.....], the minimum is ..... at index .....
2. Swap: swap minimum with first element of unsorted array
   - After swap: [.....]
3. Result: Now 3 elements sorted. The remaining unsorted array is: [.....]

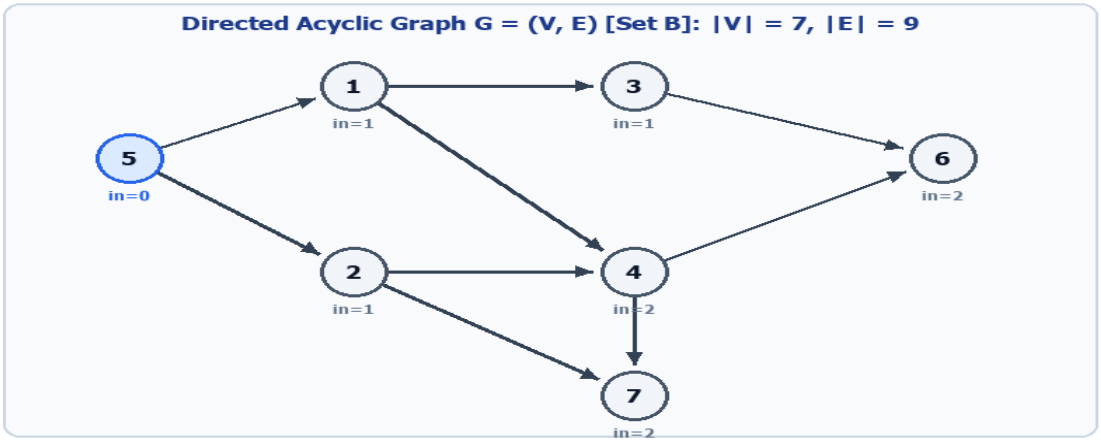
Round 4:

1. Find minimum: From unsorted array [.....], the minimum is ..... at index .....
2. Swap: swap minimum with first element of unsorted array
   - After swap: [.....]
3. Result: Now 4 elements sorted. The remaining unsorted array is: [.....]

4.2 Maximum Swaps: What is the maximum number of swaps performed by Selection Sort for N = 7? ..... swaps [2 Marks]

Question 5: Topological Sort (Kahn's Algorithm on 7-Vertex DAG) (10 Marks)

Given the Directed Acyclic Graph (DAG) shown below with |V| = 7 vertices and |E| = 9 directed edges:



5.1 In-Degree Array Table: Calculate initial in-degree for all vertices: [2 Marks]

Vertex	1	2	3	4	5	6	7
In-Degree							

5.2 Circular Queue Simulation Table: Trace Queue states as vertices are processed: [4 Marks]

Step #	Action (Dequeue / Enqueue)	Queue Contents	Front Ptr	Back Ptr
1				
2				
3				
4				
5				
6				
7				
8				

5.3 Final Topological Ordering: State the final output order of vertices: [2 Marks]

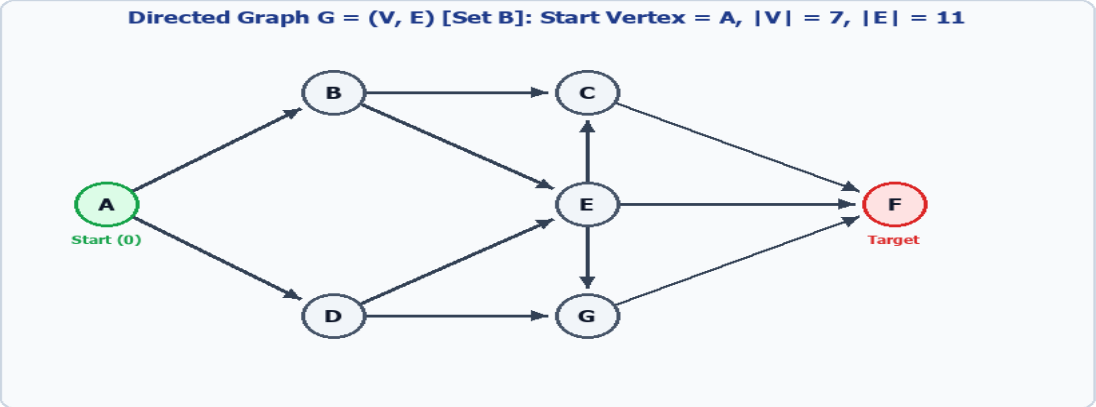
- Order: .....

5.4 Directed Cycle Detection: What happens to Kahn's algorithm if the graph contains a directed cycle? [2 Marks]

- Explanation: .....

Question 6: Unweighted Shortest Path Algorithm (BFS) (10 Marks)

Given the directed graph below with |V| = 7 vertices (A to G) and |E| = 11 edges.
Start vertex is Vertex A. Execute Breadth-First Search (BFS) to find the shortest path to all vertices:



6.1 Shortest Path Processing Table (Assignment 4 Style): [5 Marks]
(Cross out previous values when updated: Known starts F, Dist starts 999 except start vertex, Path starts 0)

Vertex (v)	Known (T / F)	Dist (d_v)	Path (p_v)
A			
B			
C			
D			
E			
F			
G			

6.2 Circular Queue Tracing: [2 Marks]

Step	Action	Queue State
1		
2		
3		
4		
5		
6		
7		

6.3 Backtracking Analysis & Shortest Path Query: [3 Marks]

- Find shortest path from Vertex A to Vertex F:
  Length (exact number of edges): ..... edges | Path Sequence: .....
- Find shortest path from Vertex A to Vertex G:
  Length (exact number of edges): ..... edges | Path Sequence: .....

(Strict rule: Length must be integer number of edges, NOT cm/inches. Path must be comma-separated, NO arrows)

Question 7: Advanced Theory, Structural Properties & Percentage Calculations (10 Marks)

Answer the following questions concisely. All calculations must be computed as exact integers or percentages (No formulas/powers).

7.1 Structural Properties (Two Properties vs Three Properties): [3 Marks]

(a) Which fundamental data structure possesses exactly two formal properties? State the data structure and define both properties:
- Data Structure: .....
- Property 1: .....
- Property 2: .....

(b) Which theoretical concept possesses exactly three formal properties? State the concept and list all three properties:
- Concept: .....
- Property 1: .....
- Property 2: .....
- Property 3: .....

(c) Which of the three properties in (b) is impossible to achieve in practice, and state the two fundamental reasons why:
- Impossible Property: .....
- Reason 1: .....
- Reason 2: .....

7.2 Graph Memory Waste Percentage Calculation: [2 Marks]

A directed graph has |V| = 8 vertices and |E| = 14 directed edges.
- Total cells in Adjacency Matrix: ..... cells.
- Number and percentage of cells containing 1 (active edges): ..... cells (..... %)
- Number and percentage of cells containing 0 (wasted space): ..... cells (..... %)
- Total memory cells required if represented as an Adjacency List (|V| + |E|): ..... cells.
- Percentage of memory saved by Adjacency List compared to Adjacency Matrix: ..... %

7.3 Complete Graph Edge Calculations: [1.5 Marks]

(a) A complete undirected graph K_{12} has |V| = 12 vertices. Calculate the total number of edges as an exact integer:
Calculation & Answer: ..... edges.

(b) A complete directed graph with 12 vertices without self-loops has: ..... directed edges.

(Strict rule: Answers must be evaluated integers, NOT mathematical formulas)

7.4 Complete Binary Tree Node Bounds: [2 Marks]

For a Complete Binary Tree where the Root is at height h = 0, calculate exact integer bounds:
- Height h = 12: Minimum nodes = ... nodes | Maximum nodes = ... nodes.
- Height h = 15: Minimum nodes = ... nodes | Maximum nodes = ... nodes.

(Strict rule: Answers MUST be computed exact integers, NOT exponential expressions like 2^{12} or 2^{15})

7.5 Instructor Exam Traps & Specific Definitions: [1.5 Marks]

(a) In graph theory, why does writing path sequence with arrows (e.g., A → B → C) result in zero marks? What is the correct format?
- Explanation & Format: .....

(b) If asked for the shortest path between two vertices in a Null Graph (V = {}, E = {}), what is the correct response?
- Answer: .....

***** END OF EXAMINATION PAPER — WISHING YOU THE BEST OF LUCK! *****